Project Methodology

What is a methodology?

A methodology is the way the team works to build the system from the start to the end. It tells us in what order we do the work (planning, analysis, design and implementation), how the team works together, and how we deal with changes. Choosing the right methodology is important because a wrong choice can make the project late or expensive.

Types of methodologies

- **1. Waterfall:** the work goes step by step. We finish one phase completely, then move to the next one, and we don't go back. It is good when the requirements are clear and will not change, and it has a lot of documentation. Its problem is that changes are difficult and the user sees the system only at the end. Example: NASA's Apollo program.
- **2. Parallel:** the project is divided into smaller parts, and different teams build them at the same time, then join them together. It is faster for big projects, but joining the parts can cause problems and delays. Example: the Boeing 787.
- **3. Iterative / RAD:** the system is built as versions (version 1, version 2, version 3). Each version goes through analysis, design and implementation, and users give their opinion on each one. RAD focuses on speed by using fast prototypes (simple models of the screens). Example: Microsoft Windows and a bank's mobile app.
- **4. Agile:** the team works in short cycles, delivers small working parts quickly, and talks face to face with the users all the time. It is very flexible when requirements change. It needs a team that works well together. Example: Spotify.

How to choose

We choose the methodology by asking some questions: Are the requirements clear or will they change? Do the users need to see the system early? How big is the team and the project? Is speed important?

Our choice

For the travel booking system we choose Agile and RAD, because the prices, offers and user needs change often, we want to deliver a first version quickly, and the users can give their feedback on the prototypes. The next page shows a case study that explains this choice.

Case Study: TravelEase Agency

The situation

TravelEase is a small travel agency. Customers call the office or come to it to book flights and hotels, and the employees write every booking on paper. The agency has 6 employees and about 50 bookings every day.

The problems

- Bookings are slow, and some of them are lost or written twice.
- Customers cannot see prices, flights or free rooms unless they call.
- Customers can only pay in cash at the office.
- Prices and offers change every week, so the information is often old.
- The manager cannot easily know how many bookings were made.

What the manager wants

A website where the customer can search for flights and hotels, book, pay online and cancel a booking. The manager wants the customers to start using a first version quickly, and he wants to ask for changes later when he sees what works.

Question

Which methodology is the best one for TravelEase?

Answer: Agile and RAD

- **Flexible:** prices and offers change every week, so the way we work must accept changes.
- **Fast:** we can give the manager a first version (search and book) in a short time.
- **User feedback:** the manager and the employees see the prototypes of the screens and tell us what to fix before we build everything.
- **Small team:** the team is small and can work together every day, which is perfect for Agile.
- **Step by step:** payment, offers and reports are added in the next versions.

Why not the others?

Waterfall is not good because the manager can't ask for changes after a phase ends, and he sees the system only at the end. Parallel is not good because it needs many teams, and this is a small project. Iterative alone is good, but RAD is faster because it uses prototypes and quick feedback.

Result

TravelEase will build the system in three versions: version 1 is search and booking, version 2 is online payment and accounts, and version 3 is offers, reports and cancellation. After each version, the users test it and give their comments.